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Data Cleaning Documentation Draft

Data Preprocessing

The dataset contains 200,000 records from 2023 to 2024, with the following columns: Order_ID, Order_Date, Customer_Name, City, State, Region, Country, Category, Sub_Category, Product_Name, Quantity, Unit_Price, Revenue, and Profit.

The first step was inspecting all column names for inconsistencies. Some columns were found to have extra spaces in their names, specifically Unit_Price, Revenue, and Profit. These were renamed to remove the extra spaces and ensure clean, consistent column headers.

Next, the data was checked for blanks across all columns. Each field was inspected to confirm that no critical values were missing, ensuring the dataset was complete before proceeding with analysis.

Following that, the data types of each column were verified and corrected where necessary. The columns Quantity, Unit_Price, Revenue, and Profit were confirmed to be in numeric format since they contain quantitative values. The Order_Date column was converted to a proper date type to support time-based analysis.

Text fields, including Customer_Name, City, State, Region, Category, Sub_Category, and Product_Name, were trimmed to remove any leading or trailing spaces that could cause inconsistencies during filtering and grouping.

Finally, duplicate records were identified and removed based on the Order_ID column, since Order_ID serves as a unique identifier for each transaction. This step ensured that no order was counted more than once in the analysis.

Data Modeling

The dataset was originally in a flat structure and was transformed into a star schema by separating attributes into dimension tables. The original sales table served as the fact table, named FactSales, which retained the columns Order_ID, Order_Date, Customer_Name, Product_Name, City, State, Region, Country, Quantity, Unit_Price, Revenue, and Profit.

Four dimension tables were created from the fact table. The date dimension, DimDate,

was created using DAX and contains the columns Date, Year, Quarter, Month Number, Month Name, Week Number, Day, and Day Name. This table supports time-based analysis such as monthly trends, quarterly comparisons, and seasonality.

The customer dimension, DimCustomer, contains Customer_Name and was derived from the distinct values in the fact table. The product dimension, DimProduct, contains Product_Name, Category, and Sub_Category to support product-level and category-level analysis. The location dimension, DimLocation, contains City, State, Region, and Country to enable geographic filtering and regional comparisons.

Relationships were then established in the Model view by connecting FactSales to each dimension table. Order_Date was linked to DimDate, Customer_Name to DimCustomer, Product_Name to DimProduct, and City to DimLocation. This structure improved filtering, readability, and overall analytical performance of the dashboard.

Metrics

The following key performance indicators (KPIs) were created to summarize overall business performance. Total Revenue was calculated by summing all values in the Revenue column. Total Profit was calculated by summing all values in the Profit column.

Total Orders was determined by counting the distinct values in the Order_ID column to ensure each order is counted only once. Total Quantity was calculated by summing all values in the Quantity column. Average Order Value was computed by dividing Total Revenue by Total Orders, representing the average amount generated per transaction.

Profit Margin was calculated by dividing Total Profit by Total Revenue, expressing how much profit is earned for every dollar of revenue generated. For time-based analysis, Year-to-Date Revenue and Year-to-Date Profit were also created to track cumulative performance within a given year.

To support trend comparison, additional time-intelligence measures were created. Prior Revenue Period was calculated by shifting the Total Revenue back by one year using DATEADD, allowing comparison of the current revenue against the same period in the previous year.

Revenue Growth Percentage was then derived by dividing the difference between Total Revenue and Prior Revenue Period by the Prior Revenue Period, expressing how much revenue has grown or declined relative to the prior year. Prior Period Orders was calculated by shifting Total Orders back by one month, enabling a month-over-month comparison of order volume.

Identification of Key Metrics and Measures

The key metrics and measures were selected to evaluate overall sales performance, profitability, efficiency, and growth trends from 2023 to 2024. These metrics are designed to support business decision-making by identifying performance drivers, monitoring financial health, and assessing growth sustainability.

Performance metrics include Total Revenue, Total Profit, Total Orders, and Total Quantity. These serve as core indicators of business performance by summarizing total income, earnings, transaction volume, and items sold.

To enable performance comparison over time, additional time-based measures were created. Revenue Previous Period was used to compare current revenue against historical values, while Revenue Growth Percentage measures the rate of increase or decrease in revenue over time. Prior Period Orders was also included to compare changes in order volume across periods.

Efficiency metrics include Average Order Value and Profit Margin. Average Order Value measures the average revenue generated per transaction, while Profit Margin evaluates how efficiently revenue is converted into profit.

Growth metrics focus on changes in performance over time, particularly through Revenue Growth Percentage, which highlights whether business performance is improving or declining compared to previous periods.

Analytical measures were also used to evaluate performance across different dimensions such as region, product, and other attributes. These measures help identify top-performing segments and provide deeper insights into the key drivers of revenue and profitability.

Overall, these metrics were structured to support both operational and strategic decision-making by enabling a clear understanding of revenue performance, cost efficiency, and growth trends, helping stakeholders assess both short-term outcomes and long-term business sustainability.

Sales Performance Overview (2023 - 2024)

Total Revenue

Big Number #

vs Prior Period

Total Orders

Big Number #

vs Prior Period

Average Order Value

Big Number #

Revenue / Orders

Total Profit

Big Number #

Profit Margin %



Top 5 Countries

Revenue by Sales Channel

Profit Margin by Item Type